

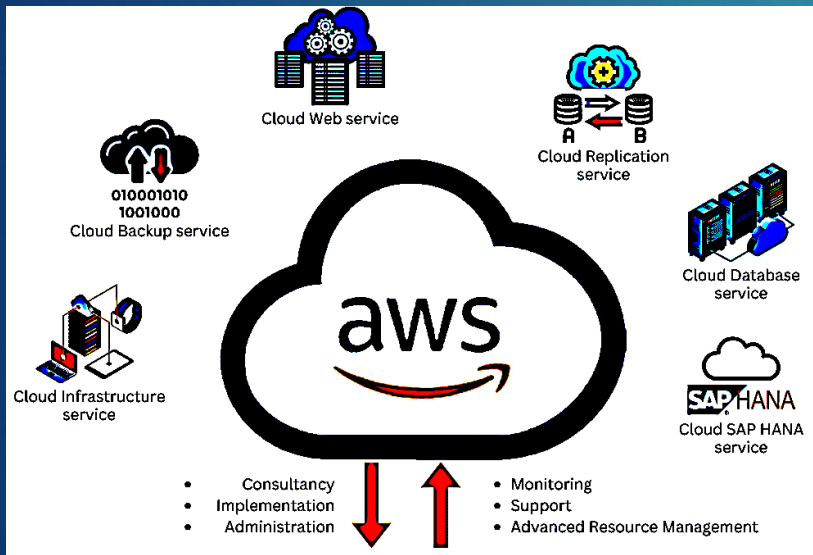
CLOUD COMPUTING

AWS-SERVICES



PROF. K. ADISESHA

AWS Case Study



AWS Services



AWS Compute



AWS Storage



AWS Database



AWS Networking

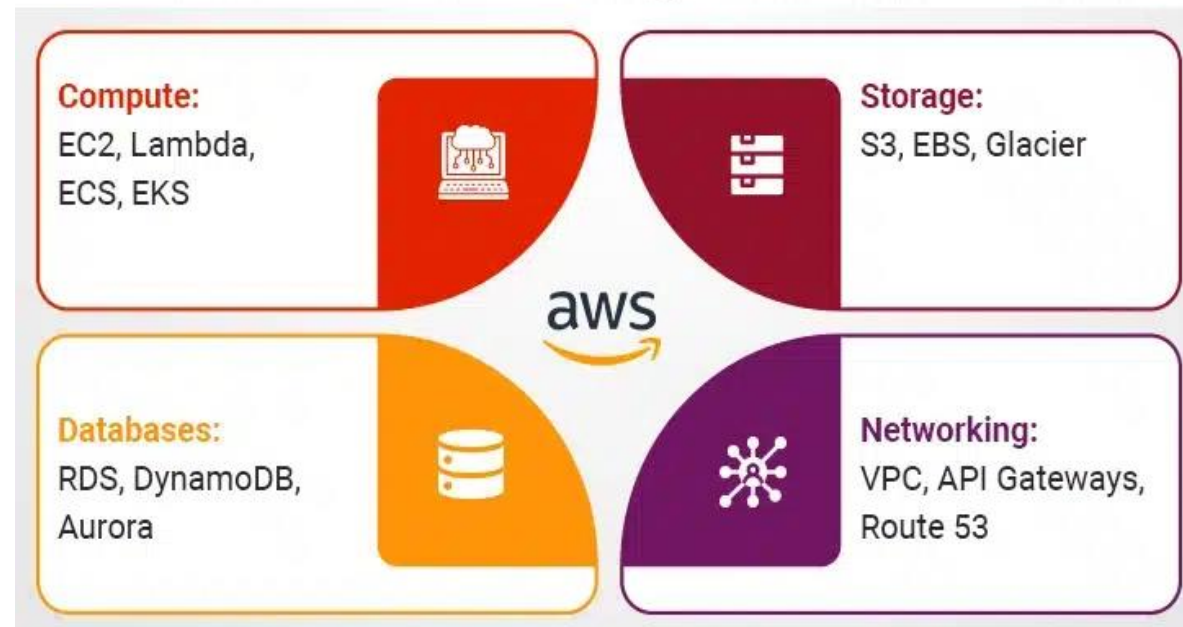
AWS Cloud Services

Amazon AWS:

Amazon Web Services (AWS) provides a comprehensive suite of services across various categories, including networking and content delivery, auto scaling, load balancing, storage, and database services.

- *The AWS Cloud spans 120 Availability Zones within 38 Geographic Regions*
- *These services are foundational for building scalable, reliable, and secure applications in the cloud.*
- [AWS Services link](#)

Core AWS Services for Building Cloud-Native Solutions



AWS Cloud Services

AWS Global Infrastructure:

AWS global infrastructure is the most secure, reliable, and extensive cloud, providing diverse infrastructure solutions for running your applications.

- *The AWS Cloud spans 120 Availability Zones within 38 Geographic Regions, with announced plans for 10 more Availability Zones and 3 more AWS Regions.*
- *AWS Region & Availability Zone*
- *AWS Edge Locations*
- *AWS Local Zones*
- *AWS Wavelength*
- *AWS Outpost*



AWS Cloud Services

Amazon AWS Services:

AWS services are the building blocks for cloud computing, including over 200 products for compute, storage, databases, and networking, all delivered on-demand over the internet.

➤ *Major services allowing businesses to rent IT infrastructure and pay for what they use include:*

- ❖ *Amazon EC2 for virtual servers*
- ❖ *Amazon VPC for Networks*
- ❖ *Amazon S3 for object storage*
- ❖ *Amazon RDS for relational databases*



Amazon EC2



Amazon S3



AWS Aurora



DynamoDB



Amazon RDS



Amazon Lambda



Amazon VPC



Amazon CloudFront



Elastic Beanstalk



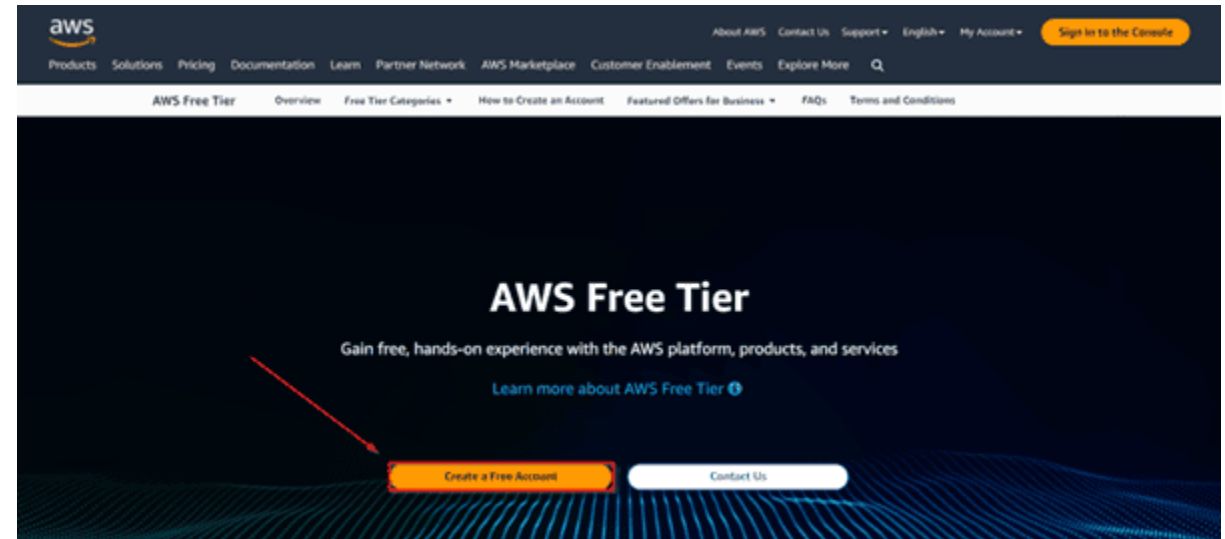
Auto-scaling

AWS Cloud Services

Creating a Free Tier Account:

The AWS Free Tier is mainly designed to give hands-on experience with AWS Cloud Services for customers free of cost for 6 Months using <https://aws.amazon.com/free/>

- *First, open your web browser and navigate to the AWS Free Tier Page.*
- *Create a Free Account*
- *Verify your email address.*
- *Password: Type your Password*
- *Captcha: Enter the given security check*
- *Payment and PAN information: fill in your card info and billing address and click on Secure Submit.*

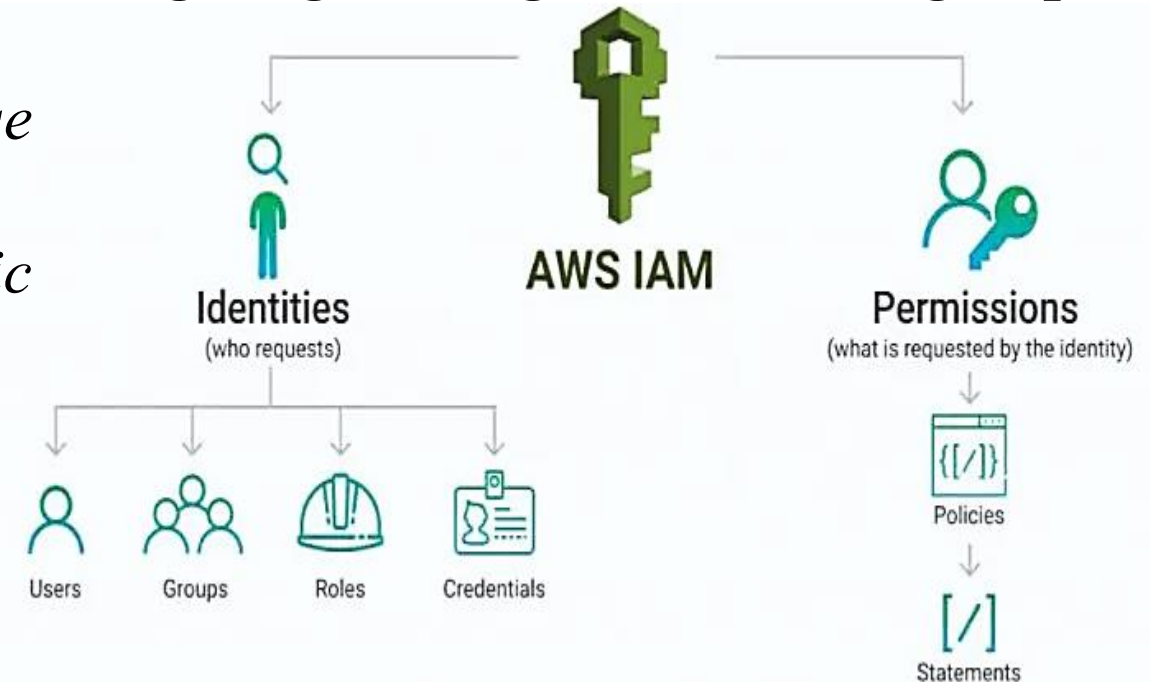


AWS Cloud Services

AWS Identity & Access Management (IAM):

Identity and Access Management (IAM) by navigating through user and group management.

- *AWS IAM is the heart of AWS security because it empowers you to control access by:*
- ❖ *Creating users and groups, assigning specific permissions and policies to specific users*
- ❖ *Managing Root Access Keys*
- ❖ *Setting up MFA Multi-Factor authentication,*
- ❖ *Additional security, and so much more.*



AWS Cloud Services

AWS IAM—Key Features:

We should consider IAM as the initial move towards making sure about all your AWS administrations and assets:

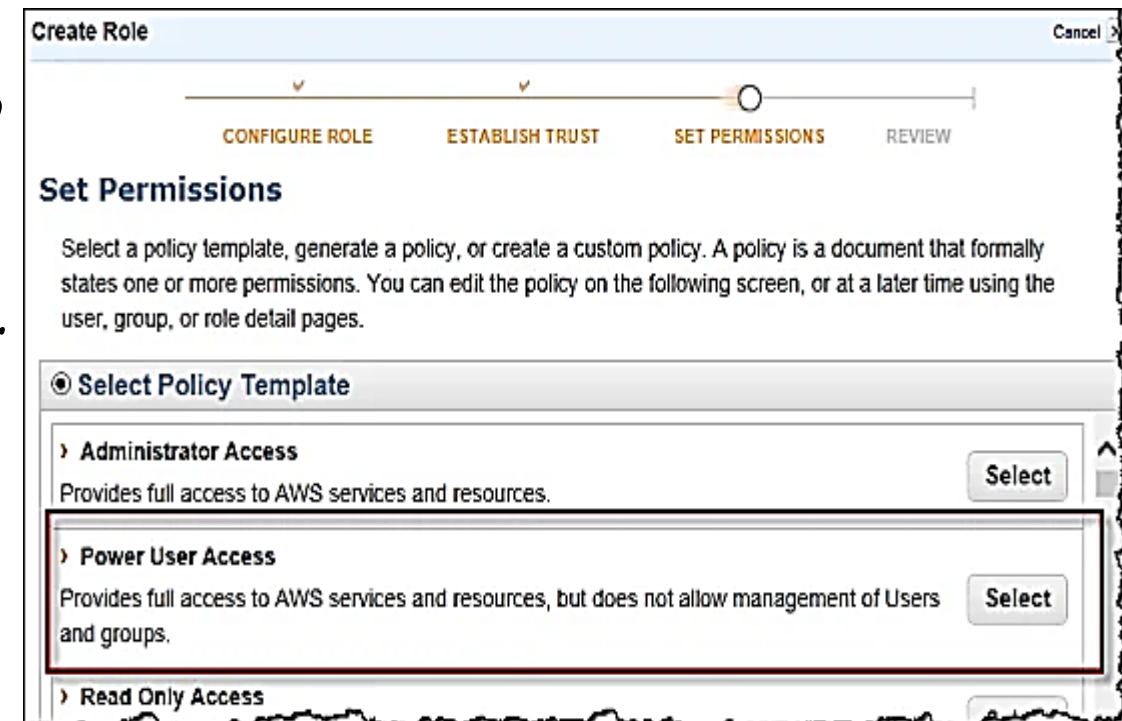
- *Confirmation: AWS IAM issue and empower verification for assets, individuals, administrations, and applications inside your AWS account.*
- *Approval: Access to the executives or approval in IAM is made of two essential segments: Policies and Permissions.*
- *Fine-grained consents: Business group in your association admittance to charging data, yet in addition need to permit the engineering group full admittance to the EC2 administration.*
- *Common admittance to AWS accounts: Most associations have more than one AWS account, and now and again need to designate access between them.*

AWS Cloud Services

IAM Power User:

Identity and Access Management (IAM) by navigating through user and group management.

- *Become an IAM power user by delving into advanced features.*
- *Learn to create and manage policies, roles, and permissions, gaining mastery over nuanced access control scenarios.*



AWS Compute Services

AWS Compute:

AWS offers services to build and manage your network, deliver content with low latency, and secure connections.

- *Amazon Elastic Compute Cloud (EC2): Provides resizable virtual servers (instances) in the cloud with customizable CPU, memory, and storage.*
- *Amazon Elastic Beanstalk(EBS): A service for deploying and managing web applications, automating tasks like provisioning and scaling.*
- *AWS Lambda: A serverless compute service that runs code in response to events without requiring you to provision or manage servers.*

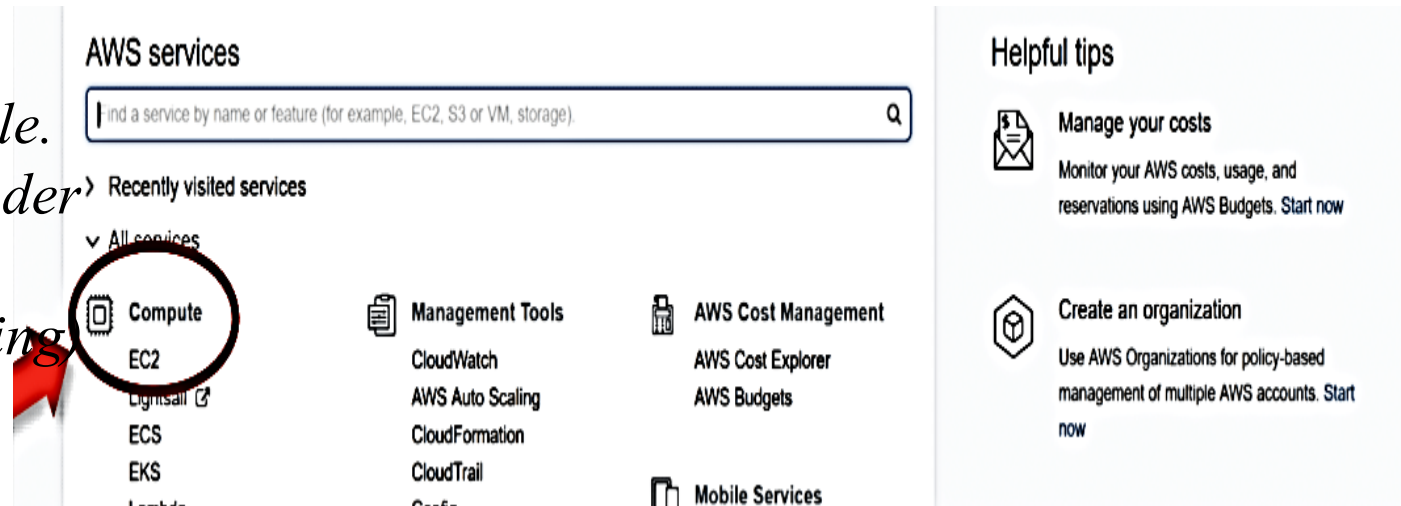


AWS Compute Service

Amazon Elastic Compute Cloud (EC2):

Amazon Elastic Compute Cloud (Amazon EC2) provides on-demand, scalable computing capacity in the AWS Cloud.

- *Amazon EC2 reduces hardware costs so you can develop and deploy applications faster.*
- *Login and Navigate to EC2 Dashboard*
- *Log in to your AWS Management Console.*
- *From the Services menu, choose EC2 under the Compute section.*
- *Under Resources, click Instances (Running) to view running instances (if any).*
- *Click Launch Instance.*

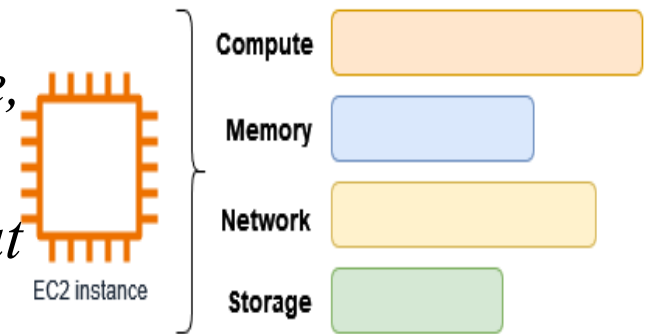


AWS Compute Service

Amazon Elastic Compute Cloud (EC2):

Amazon Elastic Compute Cloud (Amazon EC2) provides on-demand, scalable computing capacity in the AWS Cloud.

- *Amazon EC2 reduces hardware costs so you can develop and deploy applications faster.*
- *Amazon Machine Images (AMIs): Preconfigured templates for your instances that package the components you need for your server.*
- *Instance types: Various configurations of CPU, memory, storage, networking capacity, and graphics hardware for your instances.*
- *Instance store volumes: Storage volumes for temporary data that is deleted when you stop, hibernate, or terminate your instance.*

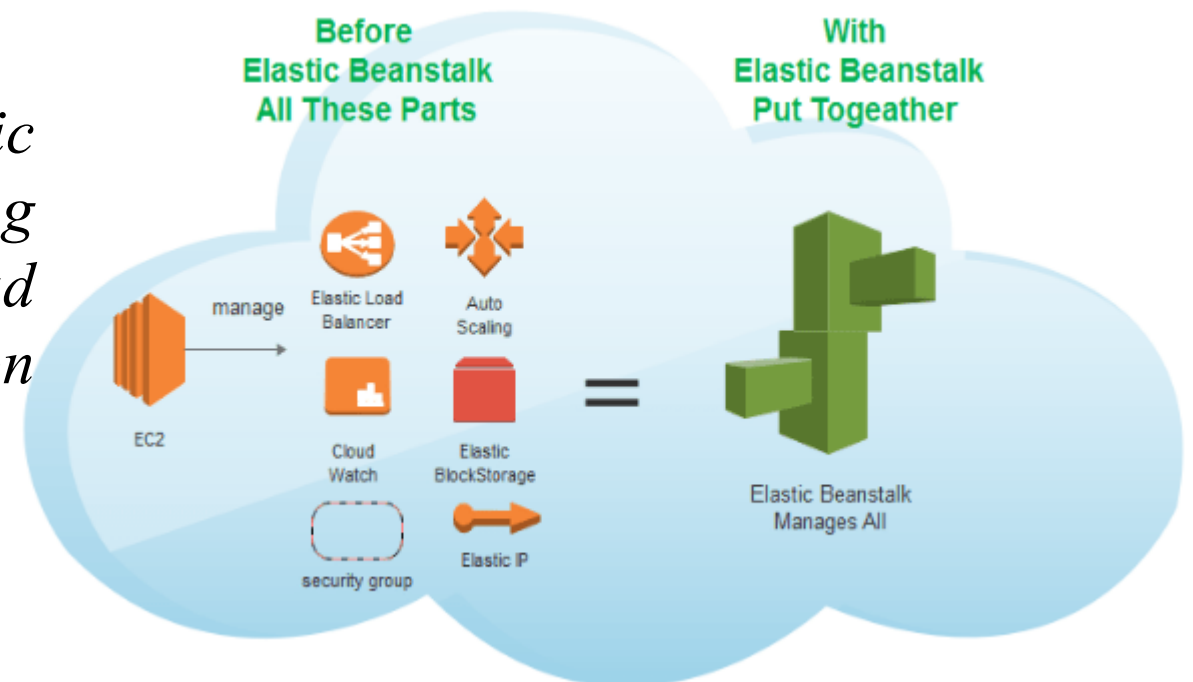


AWS Compute Service

Amazon Elastic Beanstalk:

Amazon Elastic Beanstalk is a Platform-as-a-Service (PaaS) offering from Amazon Web Services (AWS) that simplifies the deployment, management, and scaling of web applications and services.

- *You only need to upload your code to Elastic Beanstalk, and it will take care of everything else, including capacity provisioning, load balancing, auto-scaling, and application health monitoring.*



AWS Compute Service

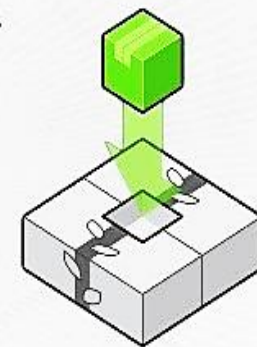
Amazon Elastic Beanstalk:

Amazon Elastic Beanstalk is the most straightforward and time-efficient method of deploying your application on AWS using the AWS Management Console, a Git repository, or an IDE.

- *A service from AWS for orchestration.*
- *Is employed to scale and deliver web applications and services.*
- *Deploying your application to AWS in the quickest and simplest manner possible.*
- *It handles application health monitoring, load balancing, deployment, capacity provisioning, and auto-scaling.*
- *Managing AWS resources is completely up to you.*

AWS Elastic Beanstalk (EB)

- Easily deploy, monitor, and scale three-tier applications
- Infrastructure provisioned and managed by EB
 - you maintain complete control.
- Preconfigured application containers
 - easily customizable.
- Support for these platforms:

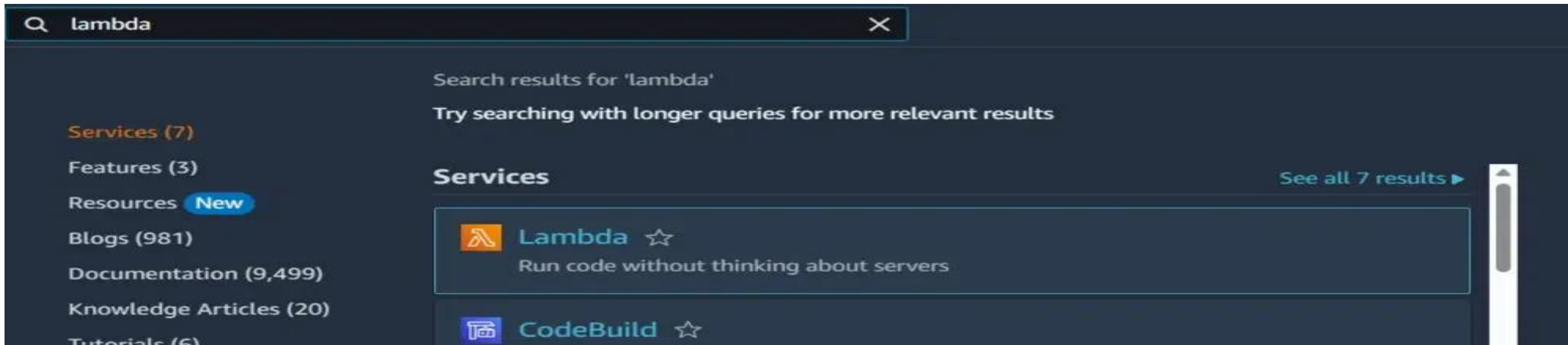


AWS Compute Service

AWS Lambda:

AWS Lambda is a serverless compute service provided by Amazon Web Services (AWS) that allows users to run code without provisioning or managing servers.

- *AWS Lambda cloud service that runs your code only when needed. You upload your code, and AWS takes care of everything else like scaling, security, and availability.*



AWS Compute Service

AWS Lambda:

AWS Lambda is a serverless compute service provided by Amazon Web Services (AWS) that allows users to run code without provisioning or managing servers.

- *AWS automatically handles the underlying infrastructure, including scaling, patching, and maintenance.*
- *Scalability: Automatically scales computing based on the rate of incoming requests.*
- *No server management: Do not require to manage servers and address optimization issues.*
- *Cost-efficient: Pay only for the usage and compute time consumed to run codes.*
- *Event-driven: Responds to other AWS services events such as DynamoDB, S3, and Kinesis.*

AWS Storage Services

Storage:

AWS provides scalable, durable, and secure storage for various data types.

➤ *Object Storage:*

❖ *Amazon S3*

➤ *Block Storage:*

❖ *Amazon EBS*

➤ *File Storage:*

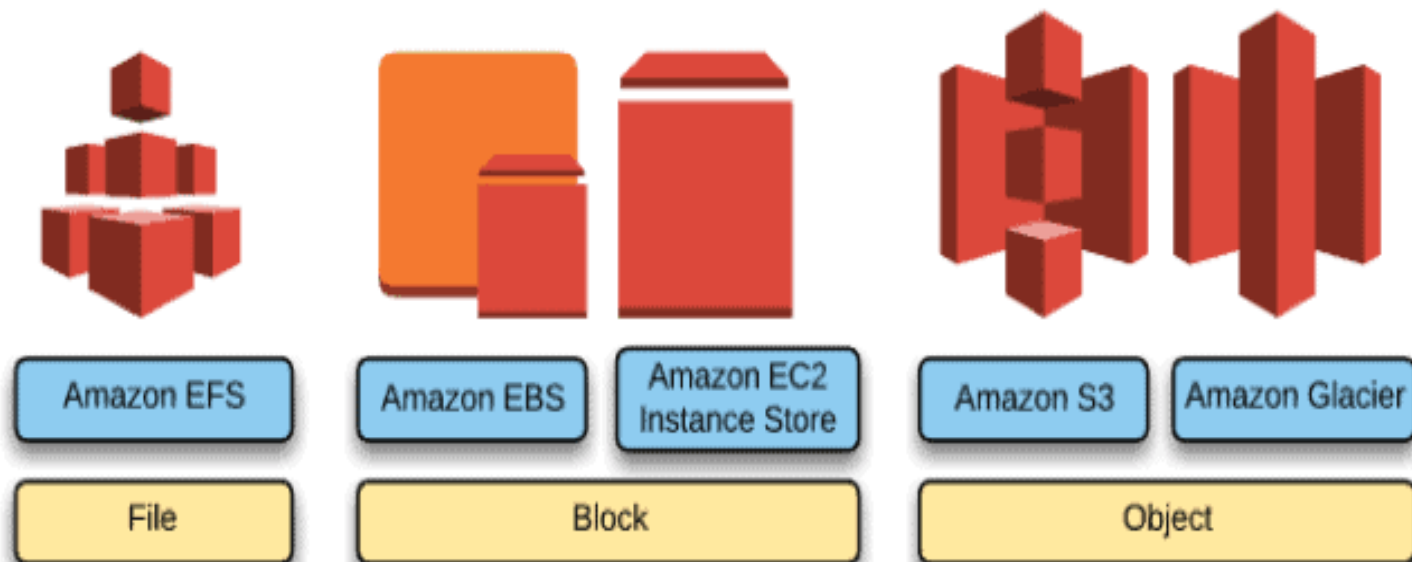
❖ *Amazon EFS*

❖ *Amazon FSx*

➤ *Hybrid Storage:*

❖ *Amazon S3 Glacier*

❖ *AWS Storage Gateway*

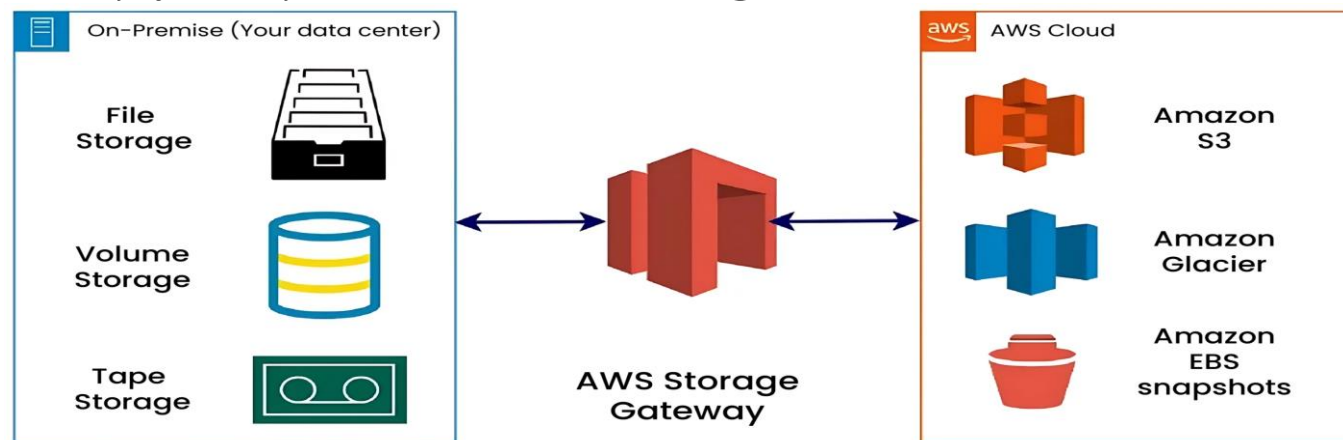


AWS Storage Services

Storage:

AWS provides scalable, durable, and secure storage for various data types include:

- *Amazon S3 for object storage,*
- *Amazon Elastic Block Store (EBS) for persistent block storage for EC2 instances*
- *Amazon Elastic File System (EFS) for a scalable file system for Linux workloads*
- *Amazon S3 Glacier for archiving,*
- *AWS Storage Gateway for hybrid cloud storage.*



AWS Storage Services

Storage:

AWS provides scalable, durable, and secure storage for various data types.

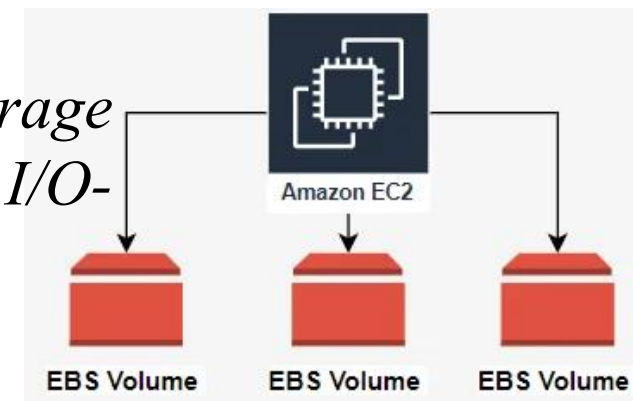
➤ *Object Storage:*

❖ *Amazon S3: Provides unlimited object storage in the cloud. Data is stored in containers called buckets and is suitable for use cases like data lakes and website hosting.*



➤ *Block Storage:*

❖ *Amazon EBS: Provides high-performance, scalable block storage for Amazon EC2 instances. It is ideal for mission-critical and I/O-intensive applications.*



AWS Storage Services

Storage:

AWS provides scalable, durable, and secure storage for various data types.

➤ *File Storage:*

- ❖ *Amazon EFS: A serverless, fully elastic file storage service that automatically grows and shrinks as you add or remove files, requiring no manual management.*
- ❖ *Amazon FSx: A managed service for high-performance, feature-rich file systems like FSx for Windows File Server and Lustre.*

➤ *Hybrid Storage:*

- ❖ *Amazon S3 Glacier: A secure, durable, and extremely low-cost storage service for data archiving and long-term backup.*
- ❖ *AWS Storage Gateway: Connects on-premises software appliances with cloud-based storage, providing seamless integration.*

AWS Storage Services

Storage:

AWS provides scalable, durable, and secure storage for various data types.

- *Amazon S3 (Simple Storage Service): An object storage service offering industry-leading scalability, data availability, security, and performance.*
- *Amazon EBS (Elastic Block Store): Provides persistent block storage volumes for use with Amazon EC2 instances.*
- *Amazon EFS (Elastic File System): A scalable, elastic, cloud-native NFS file system for use with AWS Cloud services and on-premises resources.*
- *Amazon S3 Glacier: A secure, durable, and extremely low-cost storage service for data archiving and long-term backup.*
- *AWS Storage Gateway: Connects on-premises software appliances with cloud-based storage, providing seamless integration.*

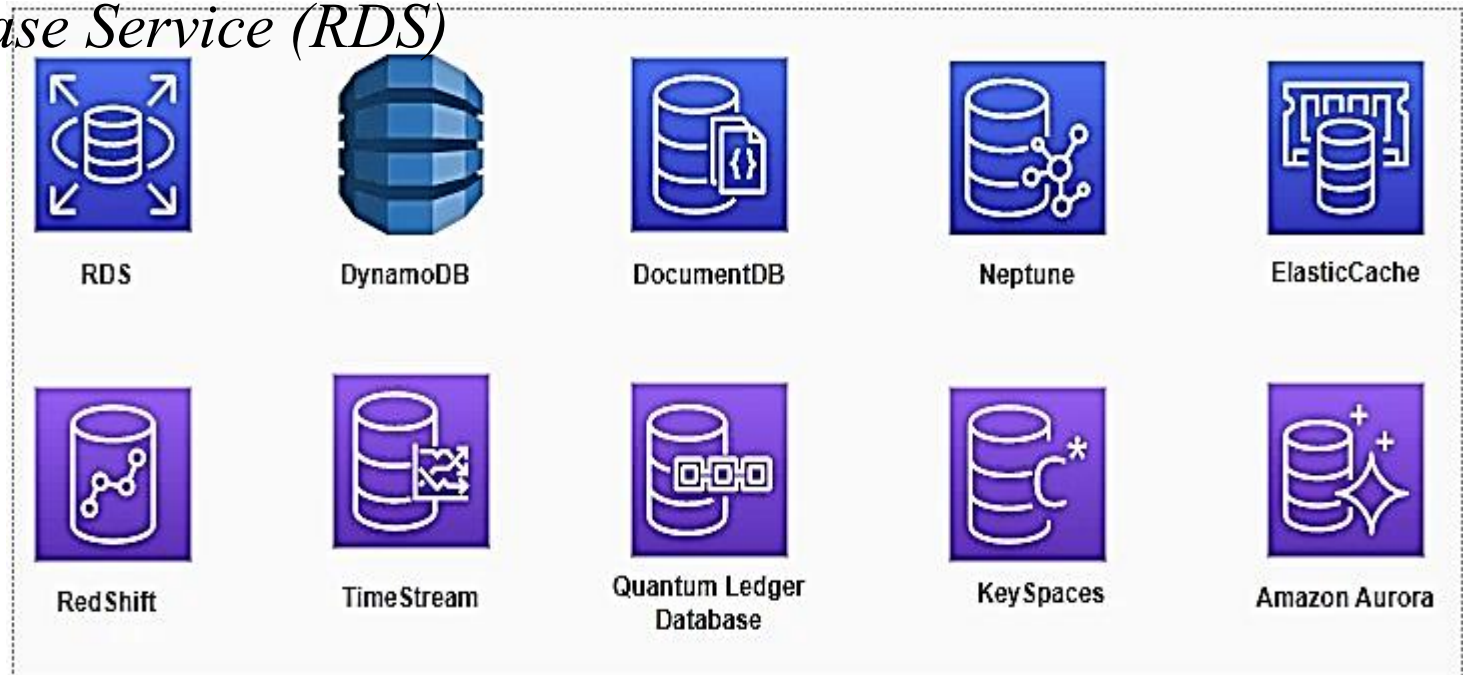
AWS Database Services

Database Services:

AWS offers a selection of database services for different application needs.

➤ *These include:*

- ❖ *Amazon Relational Database Service (RDS)*
- ❖ *Amazon DynamoDB*
- ❖ *Amazon Aurora*
- ❖ *Amazon ElastiCache*
- ❖ *Amazon Neptune*
- ❖ *Amazon Redshift*
- ❖ *Amazon TimeStream*



AWS Database Services

Database Services:

AWS offers a selection of database services for different application needs.

- *Amazon RDS (Relational Database Service): Makes it easy to set up, operate, and scale a relational database in the cloud, supporting various database engines like MySQL, PostgreSQL, Oracle, SQL Server, and Amazon Aurora.*
- *Amazon Aurora: A MySQL and PostgreSQL-compatible relational database built for the cloud, combining the performance and availability of traditional enterprise databases with the simplicity and cost-effectiveness of open-source databases.*
- *Amazon DynamoDB: A fast and flexible NoSQL database service for all applications that need consistent, single-digit millisecond latency at any scale.*
- *Amazon Redshift: A fully managed, petabyte-scale data warehouse service.*
- *Amazon ElastiCache: A fully managed in-memory data store compatible with Redis and Memcached for caching and real-time data processing.*

AWS Cloud Services

Networking and Content Delivery:

AWS offers services to build and manage your network, deliver content with low latency, and secure connections.

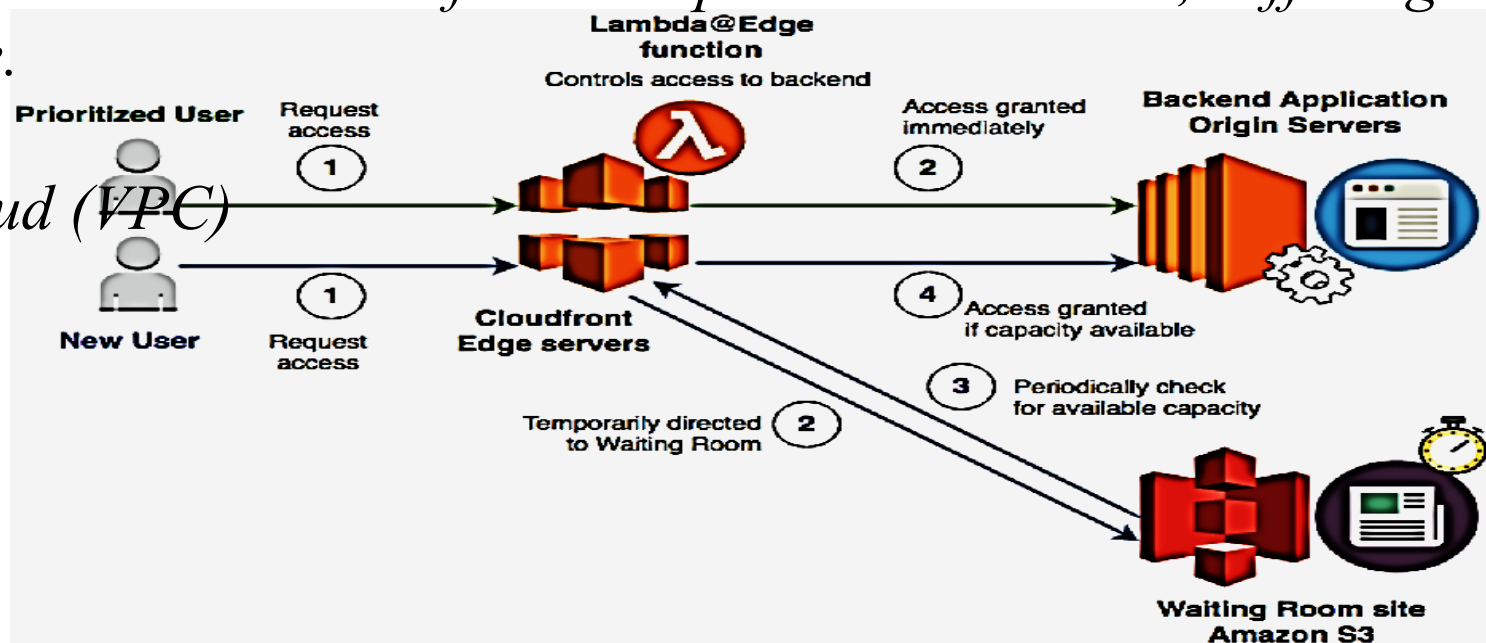
- *Establishes a dedicated network connection from on-premises to AWS, offering consistent network performance.*
- *These include:*
 - ❖ *Create a Custom Virtual Private Cloud*
 - ❖ *Work with VPC Peering Connection*
 - ❖ *AWS Elastic IP*
 - ❖ *Establish a Client Side VPN*
 - ❖ *Register a GoDaddy domain & Map using Route53*

AWS Content Delivery Services

Networking and Content Delivery:

AWS offers services to build and manage your network, deliver content with low latency, and secure connections.

- *Establishes a dedicated network connection from on-premises to AWS, offering consistent network performance.*
- *These include:*
 - ❖ *Amazon Virtual Private Cloud (VPC)*
 - ❖ *Amazon Route 53*
 - ❖ *Amazon CloudFront*
 - ❖ *AWS Direct Connect*

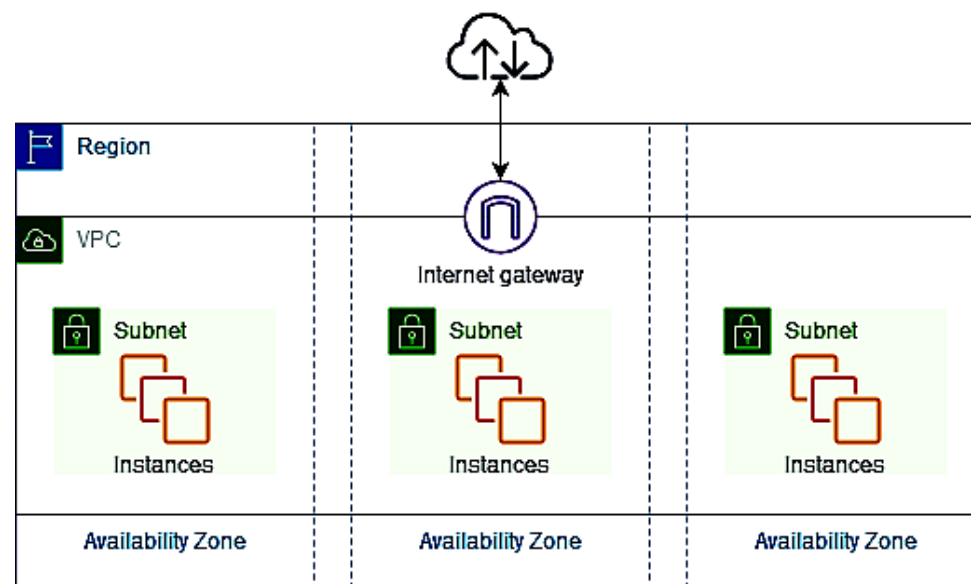


Networking and Content Delivery

Amazon Virtual Private Cloud (VPC):

Allows users to provision a logically isolated section of the AWS Cloud where they can launch AWS resources in a virtual network defined by them.

- *AWS VPC is a private subsection of AWS in which you can place AWS resources such as EC2 instances and databases.*
- ❖ *It gives all the benefits of the traditional network that you have for your own data center.*
- ❖ *It gives the benefit of scalable infrastructure in the AWS environment.*
- ❖ *It gives you complete control over your virtual network.*

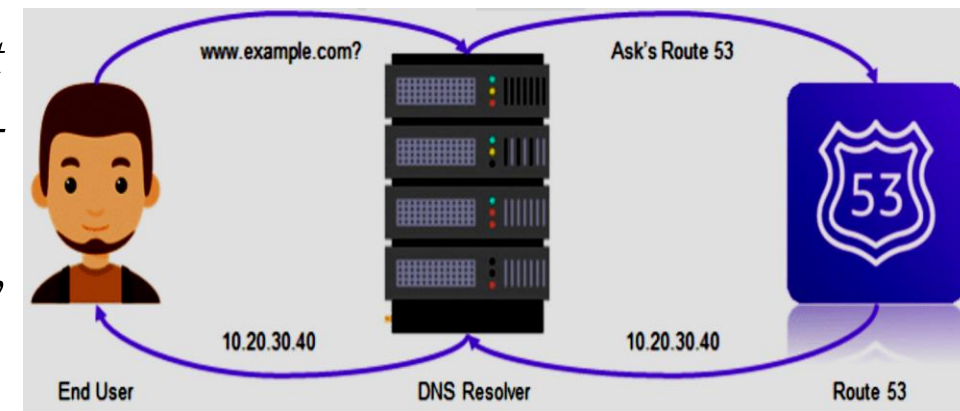


Networking and Content Delivery

Amazon Route 53:

Amazon Route 53 is a highly available and scalable cloud Domain Name System (DNS) web service provided by AWS that translates human-readable domain names into IP addresses to connect users to internet applications.

- *DNS resolution: Converts domain names (e.g., `www.example.com`) into IP addresses, enabling users to access your web applications.*
- *Traffic routing: Directs user traffic to the correct servers or endpoints, whether they are running within AWS or elsewhere.*
- *Domain registration: Allows you to register new domain names directly through the service.*

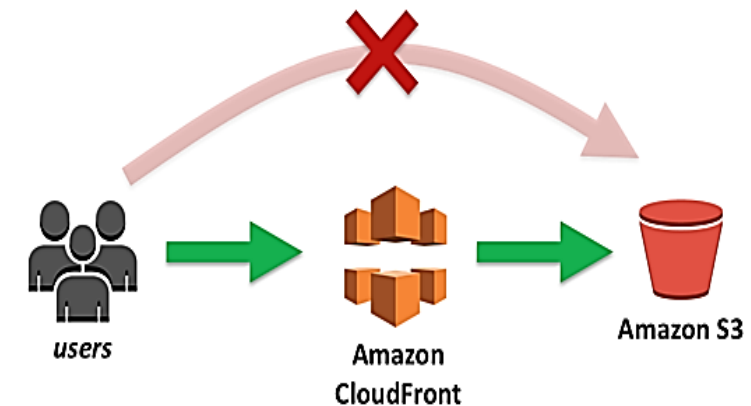


Networking and Content Delivery

Amazon CloudFront:

Amazon CloudFront is a global content delivery network (CDN) service from AWS that speeds up the distribution of static and dynamic web content, like HTML, CSS, JavaScript, and images, to end users.

- *CloudFront also enhances security by protecting against various attacks and integrating with other AWS security services.*
- *Edge locations: CloudFront uses a global network of data centers called edge locations to store copies of your content.*
- *Cache and delivery: If the content is in the cache, it's served directly from the edge location. If not, CloudFront fetches it from the origin and caches it for future requests.*

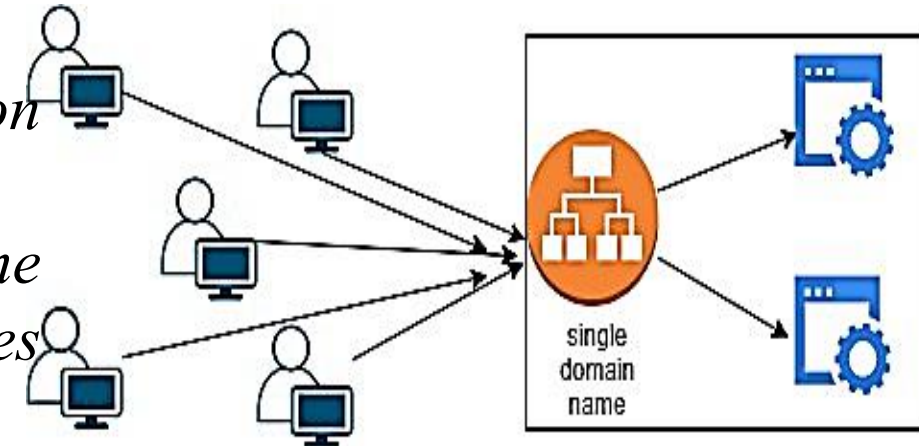


Networking and Content Delivery

Elastic Load Balancing (ELB):

Automatically distributes incoming application traffic across multiple targets, such as EC2 instances, containers, and IP addresses, in one or more Availability Zones.

- *Automatic scaling: It can automatically increase or decrease its capacity to handle varying amounts of incoming traffic.*
- *Distributes traffic: It accepts incoming traffic and forwards it to a group of registered targets.*
- *Monitors health: It performs regular health checks on the registered targets.*
- *Routes to healthy targets: If a target is unhealthy, the load balancer stops sending traffic to it and routes traffic to other healthy targets instead.*



Networking and Content Delivery

Networking and Content Delivery:

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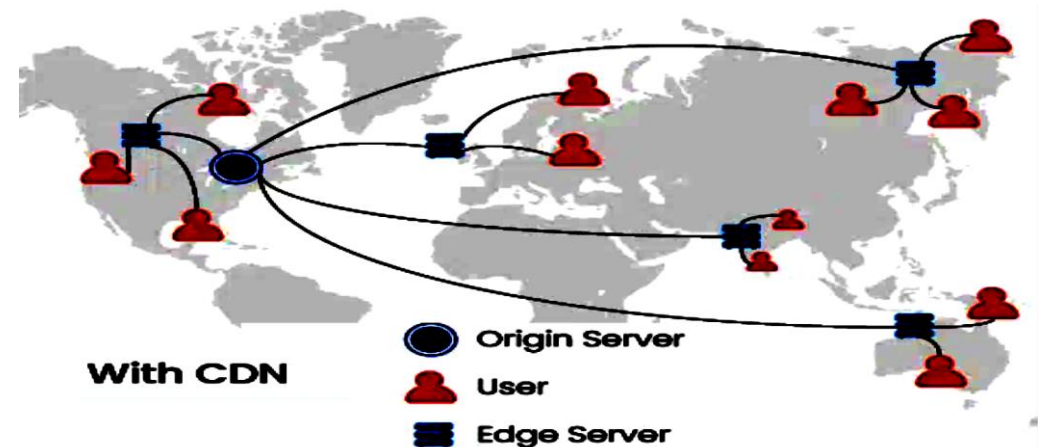
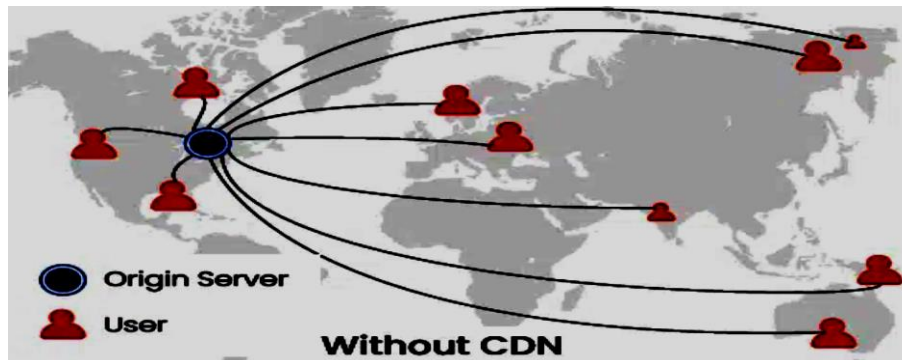
- *Amazon CloudFront: A fast content delivery network (CDN) service that securely delivers data, videos, applications, and APIs to customers globally with low latency and high transfer speeds.*
- *Amazon Route 53: A highly available and scalable cloud Domain Name System (DNS) web service.*
- *AWS Direct Connect: Establishes a dedicated network connection from on-premises to AWS, offering consistent network performance.*

Content Delivery Network(CDN)

Networking and Content Delivery:

AWS offers services to build and manage your network, deliver content with low latency, and secure connections.

- *When a user requests content from a website with a CDN, the CDN identifies the user's location and routes the request to the nearest edge server.*
- *The edge server, which stores cached copies of the website's content, quickly delivers the requested content to the user.*

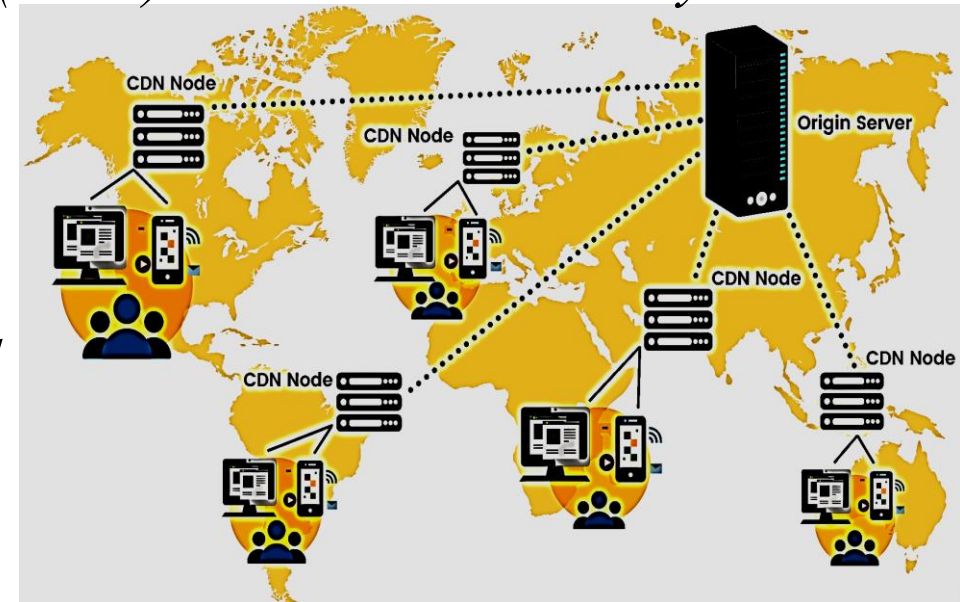


Content Delivery Network(CDN)

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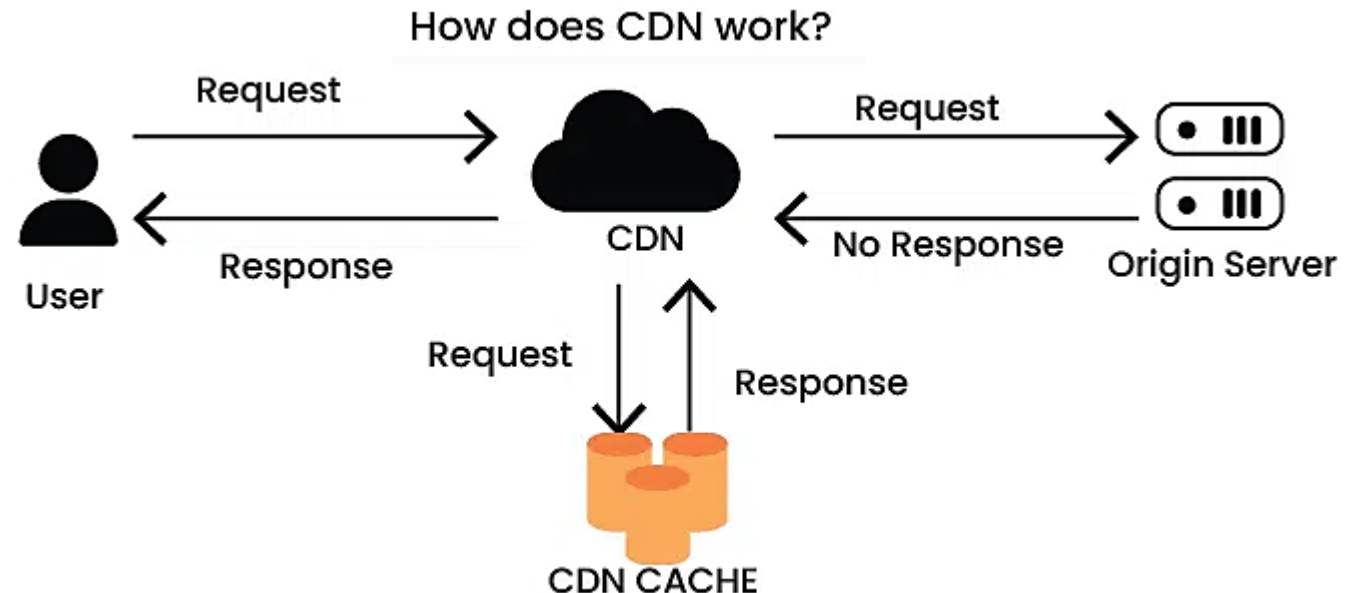
Content Delivery Network(CDN)

Content Delivery Network(CDN):

The CDN helps to offload traffic from the origin server, reducing the risk of server overload and ensuring consistent performance even during traffic spikes.

➤ *CDNs can be classified into several types based on their architecture and functionality:*

- ❖ *Public CDNs*
- ❖ *Private CDNs*
- ❖ *Peer-to-Peer (P2P) CDNs*
- ❖ *Hybrid CDNs*
- ❖ *Push CDNs*
- ❖ *Pull CDNs*



Networking and Content Delivery

Networking and Content Delivery:

CDNs can be classified into several types based on their architecture and functionality:.

- *Public CDNs: Any CDN that is accessible to everybody online is referred to as a public CDN.*
 - ❖ *These CDNs are used to swiftly and effectively provide content, including pictures, movies, and other static files, to users. For example: Cloudflare, Akamai, and Amazon CloudFront.*
- *Private CDNs: A CDN that is only utilized by one firm or organization is a private CDN.*
 - ❖ *These CDNs are used to distribute content to internal users or clients, and they are frequently set up on a private cloud or within an organization's own infrastructure.*
 - ❖ *For example: Google Cloud CDN, Netflix Open Connect.*
- *Peer-to-Peer (P2P) CDNs: These CDNs utilize peer-to-peer networking technology to distribute content directly between users, reducing reliance on centralized servers.*
 - ❖ *For example: BitTorrent, webTorrent.*

Networking and Content Delivery

Networking and Content Delivery:

CDNs can be classified into several types based on their architecture and functionality:.

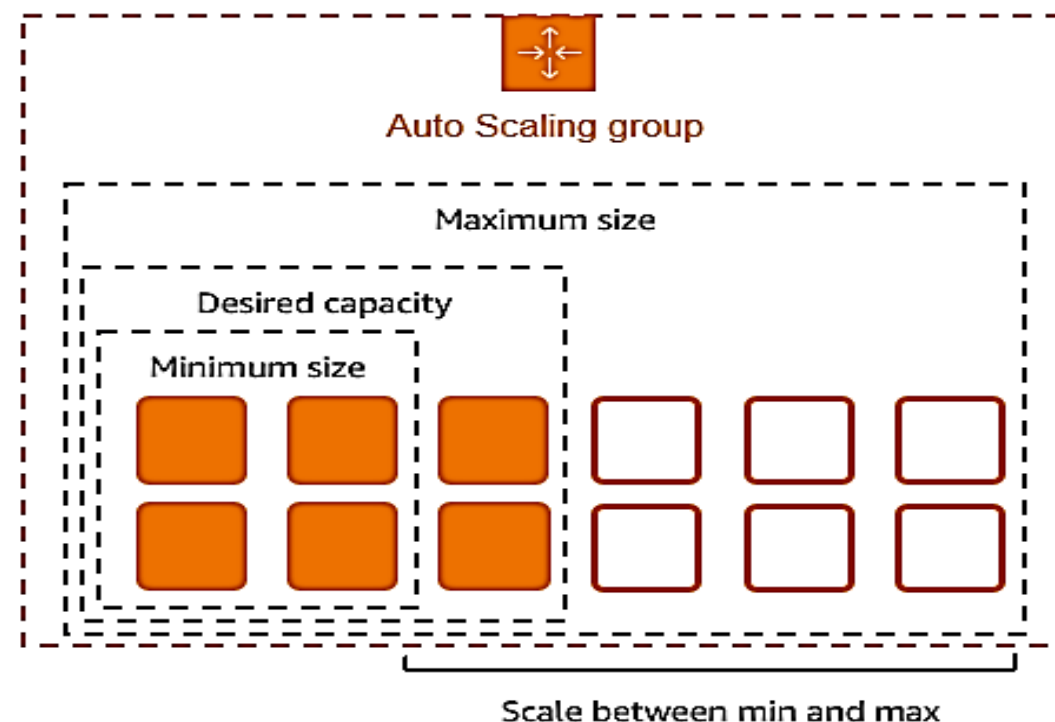
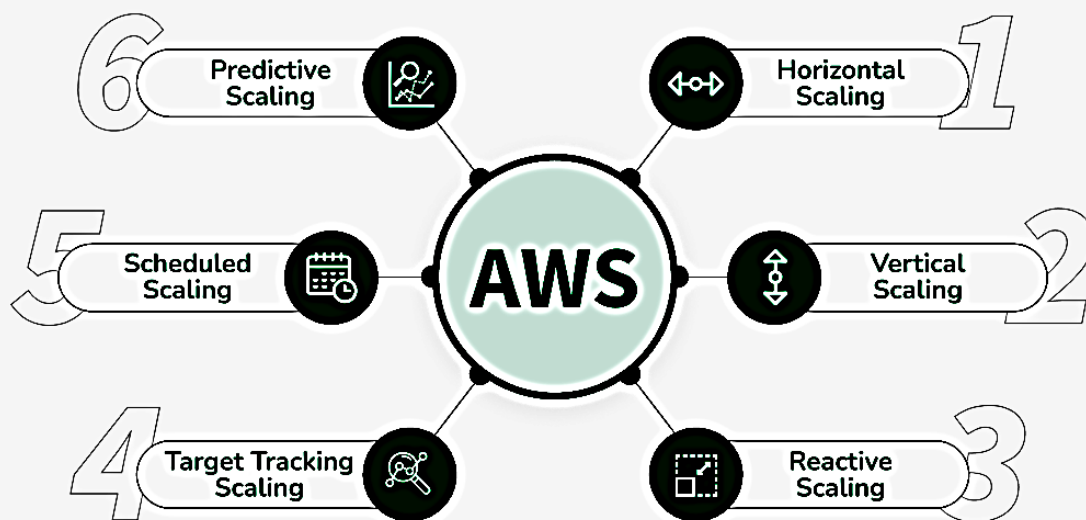
- *Hybrid CDNs: A hybrid CDN combines elements of both public and private CDNs.*
 - ❖ *In a hybrid CDN, some content is delivered using a public CDN, while other content is delivered using a private CDN. For example: Microsoft Azure CDN*
- *Push CDNs: In a push CDN, content is uploaded or "pushed" to the CDN's servers in advance of when it is needed.*
 - ❖ *Push CDNs are often used for caching large files or content that is not frequently updated.*
 - ❖ *For example: KeyCDN, CDN77*
- *Pull CDNs: In a pull CDN, content is requested or "pulled" from the CDN's servers when it is needed.*
 - ❖ *Pull CDNs are often used for delivering dynamic content, such as web pages or API responses. For example: Amazon CloudFront, Cloudflare*

AWS Cloud Services

Auto Scaling:

AWS Auto Scaling automatically adjusts application capacity to maintain performance at the lowest cost.

Types of Auto Scaling AWS



AWS Cloud Services

Auto Scaling:

AWS Auto Scaling automatically adjusts application capacity to maintain performance at the lowest cost.

- *Services include AWS Auto Scaling and Amazon EC2 Auto Scaling, utilizing various scaling policies like reactive, predictive, and scheduled scaling.*
- *AWS Auto Scaling: Monitors applications and automatically adjusts capacity to maintain steady, predictable performance at the lowest possible cost.*
- *It can scale resources across multiple AWS services like EC2, ECS, DynamoDB, and RDS.*

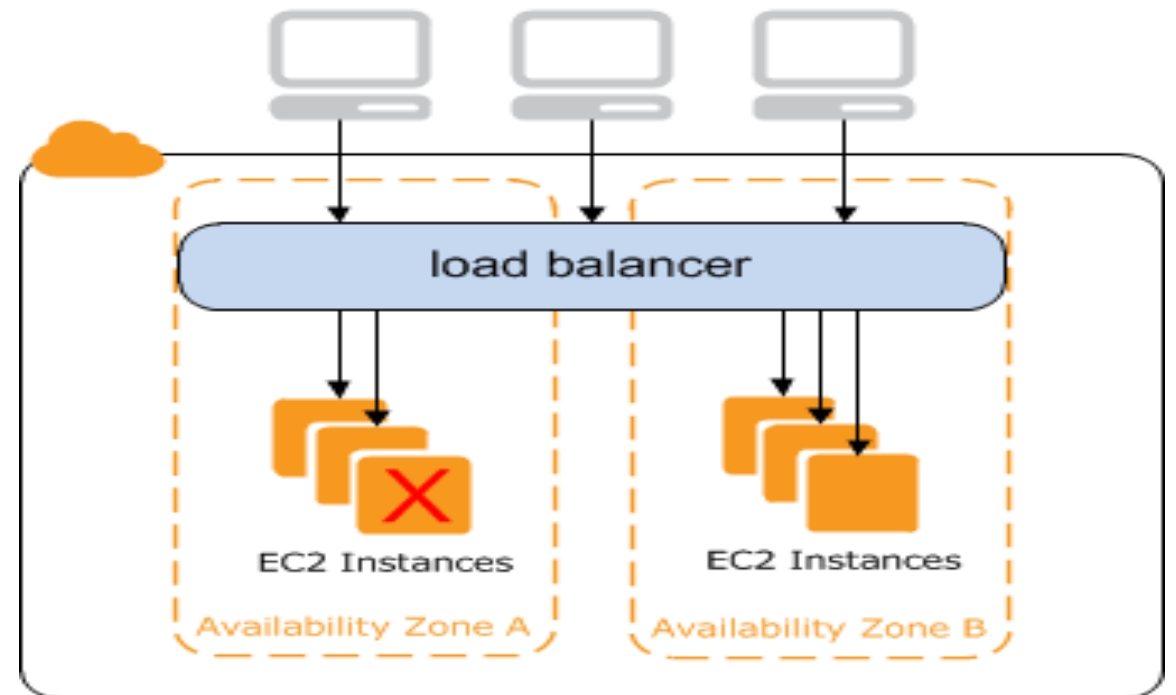
AWS Cloud Services

Load Balancing:

Elastic Load Balancing (ELB) distributes incoming traffic across targets to increase application availability and fault tolerance.

➤ *Types include:*

- ❖ *Application Load Balancer (ALB)*
- ❖ *Network Load Balancer (NLB)*
- ❖ *Gateway Load Balancer (GWLB)*
- ❖ *Classic Load Balancer (CLB)*



AWS Cloud Services

Load Balancing:

Elastic Load Balancing (ELB) distributes incoming traffic across targets to increase application availability and fault tolerance.

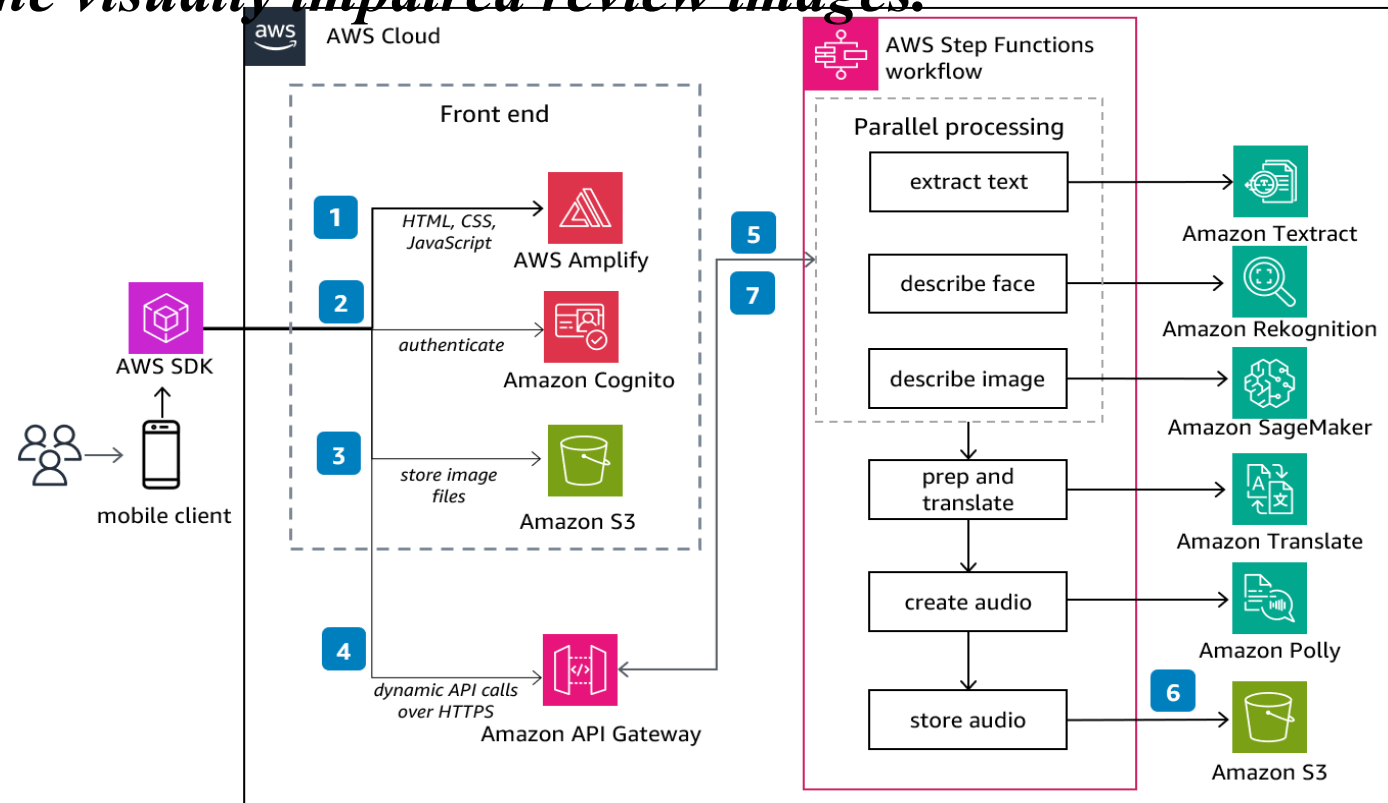
- *Types include Application Load Balancer (ALB), Network Load Balancer (NLB), Gateway Load Balancer (GWLB), and Classic Load Balancer (CLB).*
- *AWS offers different types of load balancers:*
 - ❖ *Application Load Balancer (ALB) for HTTP/HTTPS traffic*
 - ❖ *Network Load Balancer (NLB) for TCP/UDP traffic*
 - ❖ *Gateway Load Balancer (GLB) for third-party virtual appliances*

AWS Cloud Case Study

Web App for Visually impaired:

Web app uses the AWS Cloud to help the visually impaired review images.

- *Through the use of multiple artificial intelligence and machine learning (AI/ML) services, you can submit a photo and an image caption will be read back to you in a clear, natural-sounding voice in a variety of languages and dialects.*



AWS Cloud Case Study

Web App for Visually impaired:

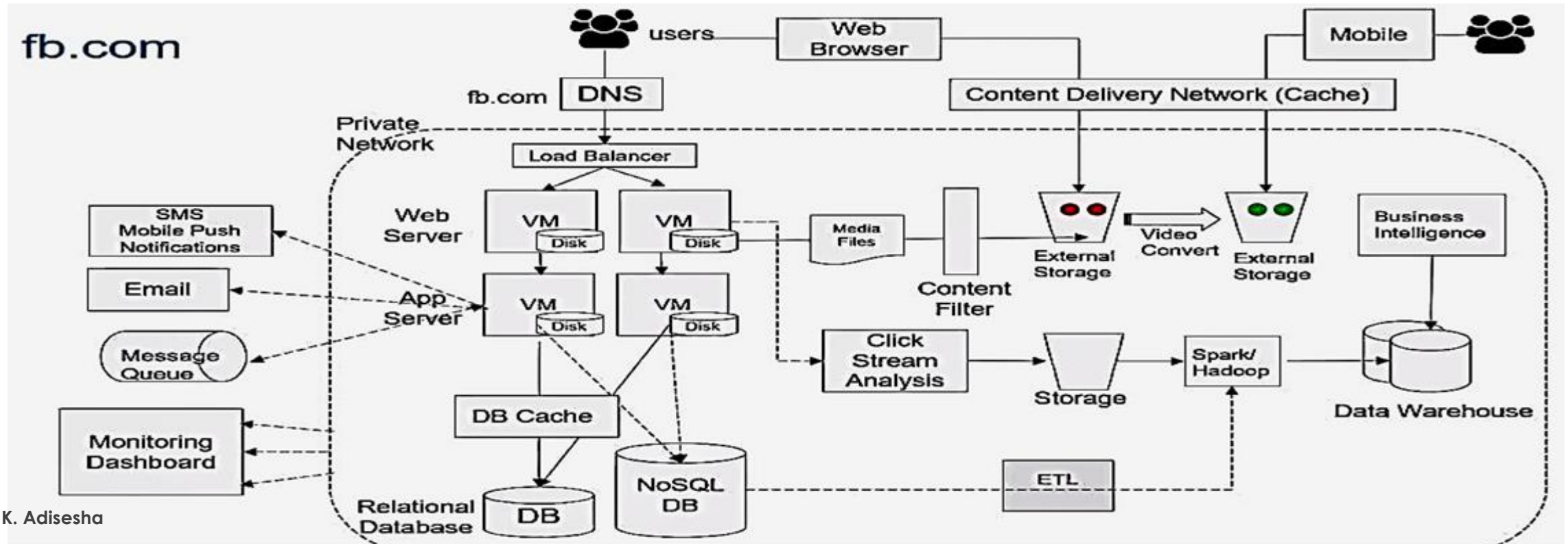
Web app uses the AWS Cloud to help the visually impaired review images.

- *AWS Amplify distributes the web app, consisting of HTML, JavaScript, and CSS, to your mobile device.*
- *The Amazon Cognito identity pool grants temporary access to the Amazon Simple Storage Service (Amazon S3) bucket.*
- *The user uploads an image file to the Amazon S3 bucket using an AWS SDK through the web app.*
- *The web app invokes the backend AI services by sending the Amazon S3 object key in the payload to Amazon API Gateway.*
- *API Gateway instantiates an AWS Step Functions workflow. The state machine orchestrates the AI/ML services Amazon Rekognition, Amazon SageMaker AI, Amazon Textract, Amazon Translate, and Amazon Polly using AWS Lambda functions.*
- *The Step Functions workflow creates an audio file as output and stores it in Amazon S3 in MP3 format.*
- *A pre-signed URL with the location of the audio file stored in Amazon S3 is sent back to your browser through Amazon API Gateway. Your mobile device plays the audio file using the pre-signed URL.*

AWS Cloud Case Study

Social Media App:

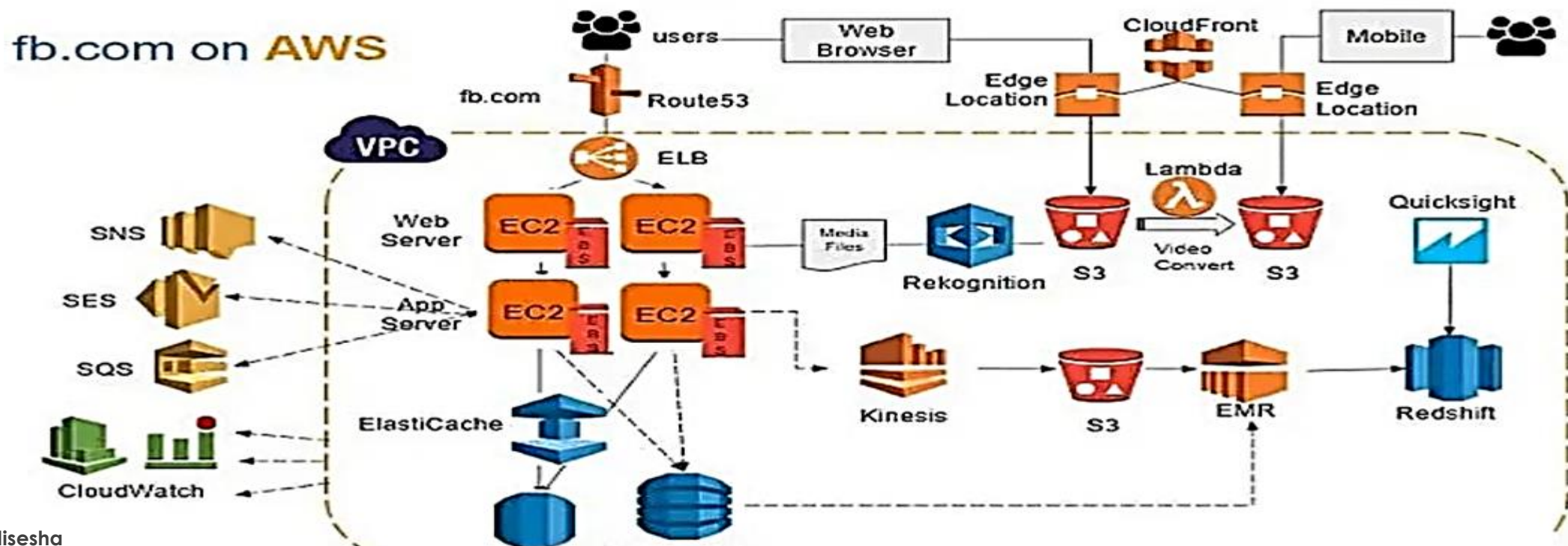
Social Media Web app using Standalone servers.



AWS Cloud Case Study

Social Media App:

Social Media Web app using AWS Cloud servers.



Discussion



Queries ?
Prof. K. Adishesha

9449081542

Thank you